

Kiran Balasundaram Kuppuraj

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EDUCATION

University of Massachusetts Amherst

Bachelor of Science in Computer Science

May 2027

GPA: 3.83/4.00

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Artificial Intelligence, SQL/NoSQL Databases, Computer Systems, JavaScript/TypeScript Programming Methodologies

Organizations: BUILD UMass Software Developer + Treasurer, PartyPilot Software Engineer (UMass AI Startup), Teaching Assistant, Undergraduate BioNLP/ML Researcher

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL

Cloud & Databases: Amazon Web Services (AWS), Azure, Firebase, Supabase, PostgreSQL, MySQL, MongoDB

Frameworks & Tools: Spring Boot, React, Node.js, Express.js, Next.js, Puter.js, Angular, Flutter, pandas, NumPy, scikit-learn, TensorFlow, Keras, PyTorch, FastAPI, Docker, Kubernetes, Postman, Git, Agile, Cursor, GitHub Copilot

EXPERIENCE

Software Engineering Intern - LeeYuen Housewares (OEM Manufacturer) *June 2025 - Aug 2025*

- Shipped a demand forecasting pipeline with **Python**, **scikit-learn**, and **AWS**, improving SKU-level sales prediction accuracy to **82%** while reducing overstock by **15%** and lowering annual storage costs by **\$20K**.
- Engineered a **React.js** + **Spring Boot** cloud-native microservice architecture, centralizing BOM, supplier pricing, and production timelines, reducing quotation turnaround by **30%** and accelerating OEM client delivery.
- Deployed **Dockerized**, **Node.js** microservices to benchmark competitor pricing across **10K+** SKUs, orchestrated on **AWS**, reducing scraping runtime by **40%** while improving accuracy of procurement insights.
- Scaled forecasting and competitor-pricing systems with **CI/CD** pipelines to support **5K+** daily users, adding monitoring and fault-tolerance that improved uptime from **95%** to **99%** and cut release cycles by **25%**.

Software Engineering Intern - Zyntra.io (Slack competitor startup) *July 2024 - Aug 2024*

- Launched a real-time, distributed messaging architecture using **React.js**, **Node.js**, **MongoDB**, and **Socket.IO**; optimized for **2K+** concurrent users in an **Agile** team with weekly code reviews and **CI/CD** deployments.
- Integrated **REST** and **GraphQL** backend APIs, with secure blob URL handling and custom state synchronization, reducing render failures and API response issues by **35%** through improved debugging and testing.
- Implemented **JWT-based authentication** with token refresh, protected routes, and session expiration logic, decreasing auth-related errors by **60%** while maintaining secure access through **Git** version control.

Data Science Intern - TruBridge Healthcare *Dec 2024 - Feb 2025*

- Executed multivariate exploratory analysis on **50K+** healthcare records via **pandas**, **NumPy**, **SQLAlchemy**; optimized ETL pipelines to handle scalable ingestion and transformation across multiple hospital systems.
- Modeled statistical relationships between hospital readmission rates and infection control KPIs using logistic regression and OLS analysis via **SciPy** and **statsmodels**, identifying high-impact predictors ($p \leq 0.01$).
- Owned end-to-end **ETL pipeline** on **AWS Lambda** + **S3**, ensuring operational excellence via error monitoring; optimized vectorized preprocessing to reduce runtime by **35%** and improve reliability for downstream analytics.

PROJECTS

StudyLensAI - 

- Built a full-stack EdTech platform with **Next.js** + **React**, enabling students to query PDFs via **AI**-powered chatbot, generate flashcards with spaced repetition, and create personalized exams through agentic workflows.
- Engineered a **RAG** system combining BM25 keyword search + vector embeddings, achieving **90%** retrieval accuracy while reducing PDF-to-embedding processing by **45%** through serverless batch optimization on **Convex**.

Kapok - 

- Collaborated on a **Flutter/Dart** disaster relief app with offline-first, enabling **200+** volunteers across **10+** zones to create teams, assign tasks via MapBox, and submit **1K+** field reports daily with role-based access control.
- Tripled** data submission rates and achieved **40%** faster sync by engineering a distributed **Firebase** backend with conflict resolution and optimized batch uploads for automated submission upon reconnection.